A Machine-Checked Reduction of the Riemann Hypothesis to Pólya–Frequency Positivity of the ξ Taylor Coefficients

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Abstract

We present a formally verified study, in Lean 4 / Mathlib, of the Laguerre–Pólya / total-positivity approach to the Riemann Hypothesis (RH). **We do not prove RH.** We prove a chain of *conditional reductions*, each checked by the Lean kernel and depending only on the foundational axioms {propext, Classical.choice, Quot.sound}. They convert RH into total-positivity statements about an explicit positive kernel and into an operator-positivity (pseudo-Hermitian) statement. The central formal statement is a *tightness theorem*. Assume three classical inputs that are not machine-checked here (Pólya–Jensen, Aissen–Schoenberg–Whitney/Edrei and the Laguerre–Pólya closure) and the classical converse. Then RH is *equivalent* to total positivity of the Toeplitz matrix of the sign-normalized ξ Taylor coefficients. The equivalence itself is classical; it is stated explicitly for ξ by Katkova. Our contribution is a machine-checked form with every external input named. It makes precise that the lone open input cannot be "postulated" as progress: doing so is logically identical to assuming RH. Along the way we identify and repair a target-function error. The error was building the program on Mathlib's entire completed zeta Λ_0, whose zeros are *not* the Riemann zeros, rather than on Riemann's ξ. We also record a moment-determinant ladder, a Turán-to-Jensen lemma in degree two, and an independent pseudo-Hermitian reduction. Numerically, we start from rigorous FLINT/Arb enclosures of the coefficients $\mu_0, \ldots, \mu_{120}$. In ball arithmetic, we certify that all 1816 contiguous Toeplitz minors of order ≤ 16 in this range are strictly positive. We also certify that (μ_n) is strictly log-concave for $n \leq 118$. These are consistency checks against known theorems, not new positivity results; the known theorems are Katkova's finite-order Pólya-frequency theorem, the Csordas–Norfolk–Varga Turán inequalities and a recent proof of infinite log-concavity. The open content is *uniformity* in the order. This version updates the June 2026 draft. It certifies the numerics, corrects their positioning against the literature, and points to an archived, axiom-audited Lean release.

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1 Introduction and scope

Disclaimer. This paper is a *formalization and reduction study*, not a proof of the Riemann Hypothesis, and it makes no claim to be one. Every implication labelled "verified" is checked mechanically; every classical theorem we rely upon but do not formalize is cited explicitly as an *external input* and is never silently assumed. The defensible contributions are:

1. a correct, machine-checked and *tight* formal version of the classical reduction of RH to a single sign-correct positivity statement;
2. the identification and repair of a target-function error that arises naturally when working with Mathlib's completed zeta;
3. an independent operator-positivity reduction;
4. reproducible, ball-arithmetic consistency checks of the reduction's premise in a finite range.

We do not claim that the equivalence, the Turán inequalities or finite-order Pólya-frequency positivity of the ξ coefficients are new; see Section 7.

Version history and relation to later work. This manuscript was drafted in June 2026 (reserved DOI `10.5281/zenodo.20574510`) and is released here as version 1.0.0. Version 1.0.0 makes four changes:

1. it replaces two-precision floating cross-checks with ball-arithmetic certification (Section 4);
2. it removes earlier claims that the computation exceeded what is known; Katkova's theorem already yields nonnegativity of every minor of order ≤ 43;
3. it records that frontier item C of the June draft is settled in the literature;
4. it ties every cited Lean theorem to an archived commit.

The subsequent development is reported separately. A formalized Fekete-type closure from strict contiguous minors to full Pólya-frequency positivity, together with effective and Jensen receptor interfaces, appears in [24]. An independent prime-defined Weil-form programme appears in [25].

The object. Riemann's entire ξ function

$$\xi(s) \;=\; \tfrac{1}{2}\, s(s-1)\, \pi^{-s/2}\, \Gamma(s/2)\, \zeta(s) \tag{1}$$

is entire, and its zeros are exactly the nontrivial zeros of ζ; RH asserts they all lie on $\operatorname{Re} s = \frac{1}{2}$. On the critical line, the real even function $\Xi(t) = \xi(\frac{1}{2} + it)$ has a Taylor expansion

$$\Xi(t) \;=\; \sum_{n\geq 0} b_n\, t^{2n}, \qquad b_n \in \mathbb{R}, \tag{2}$$

and we set the *sign-normalized sequence* $\mu_n := (-1)^n b_n$.

Two classical pillars.

- *Pólya–Jensen* [3, 7]: RH is equivalent to hyperbolicity (all roots real) of every Jensen polynomial $J_{d,n}(X) = \sum_{k=0}^{d} \binom{d}{k} b_{n+k} X^k$.
- *Aissen–Schoenberg–Whitney / Edrei* [4, 5, 6]: a real sequence is a *Pólya-frequency* (PF) sequence—all minors of the Toeplitz matrix $[a_{i-j}]$ are nonnegative—iff its generating function is $e^{\gamma z}\prod(1+\alpha_i z)/\prod(1-\beta_j z)$ with $\gamma, \alpha_i, \beta_j \geq 0$; for an entire generating function this is exactly Laguerre–Pólya membership.

Together: *RH* $\iff$ (μ_n) *is a Pólya-frequency sequence.* This consequence is classical; Katkova [13] states it explicitly for $\xi_1(z) = \xi(\frac{1}{2} + \sqrt{z}) = \sum_n \mu_n z^n$. She also proves that (μ_n) is a Pólya-frequency sequence of every finite order up to an explicit bound, using a zero-free sector theorem of Schoenberg. Her printed proof gives order 43, while her statement gives 44; we use 43. Here μ_n coincides with Katkova's coefficients and with $\lambda_n = \xi^{(2n)}(\frac{1}{2})/(2n)!$ of [20].

Pólya's positive kernel. By Pólya's representation [2], $\Xi(t) = 2\int_0^\infty \Phi(u)\cos(ut)\, du$ with $\Phi > 0$, so

$$b_n = (-1)^n \frac{2M_n}{(2n)!}, \qquad M_n := \int_0^\infty \Phi(u)\, u^{2n}\, du > 0, \qquad \mu_n = \frac{2M_n}{(2n)!} > 0. \tag{3}$$

2 The target-function correction: ξ, not Λ_0

We record an error and its repair, because catching it is part of the contribution. Mathlib's *entire* completed zeta $\Lambda_0 =$ `completedRiemannZeta`$_0$ satisfies $\Lambda =$ `completedRiemannZeta` $= \Lambda_0 - 1/s - 1/(1-s)$.

Proposition 1 (Wrong target)**.** *The zeros of Λ_0 are not the nontrivial zeros of ζ. Indeed, at a nontrivial zero ρ one has $\Lambda(\rho) = 0$ but $\Lambda_0(\rho) = \frac{1}{\rho} + \frac{1}{1-\rho} \neq 0$. On the critical line, $\Lambda_0(\frac{1}{2} + it) = (1 - 2\,\Xi(t))/(t^2 + \frac{1}{4})$, whose zeros form the level set $\{\Xi = \frac{1}{2}\}$, unrelated to RH.*

The remedy is to remove the poles of Λ by *multiplying* by $s(s-1)/2$ (which preserves zeros), recovering ξ of (1). We verify the corrected slice has the correct zero set:

Theorem 2 (`Xi_eq_zero_iff_riemannZeta`; verified)**.** *For all $t \in \mathbb{R}$, $\Xi(t) = 0 \iff \zeta(\frac{1}{2}+it) = 0$.*

Proof (mechanized). We define $\Xi(t) = -\frac{t^2+1/4}{2}\, Z(t)$ where $Z(t) = \operatorname{Re}\Lambda(\frac{1}{2} + it)$ is the verified real critical-line slice. The scalar $-\frac{t^2+1/4}{2}$ is a nonzero real, so $\Xi(t) = 0 \iff Z(t) = 0 \iff \Lambda(\frac{1}{2}+it) = 0 \iff \zeta(\frac{1}{2} + it) = 0$ (the last step using $\Lambda = \Gamma_{\mathbb{R}} \cdot \zeta$ with $\Gamma_{\mathbb{R}}(\frac{1}{2} + it) \neq 0$). All steps are checked in Lean. □

All total-positivity, Jensen, and moment results below are stated for this corrected ξ.

3 Verified reductions

Throughout, "verified" means the Lean theorem reduces to {propext, Classical.choice, Quot.sound}, confirmed by `#print axioms`. Definitions of type `Prop` that encode classical theorems we do not formalize are *named external inputs*; they are never asserted, only taken as explicit hypotheses.

3.1 The Pólya-frequency / Toeplitz route

Definition 3 (`XiToeplitzTotalPositive`)**.** The sign-normalized sequence (μ_n) is *Pólya-frequency total positive* if every minor of the Toeplitz matrix $[\mu_{i-j}]$ (over all strictly monotone row and column index selections) is nonnegative.

Proposition 4 (Low-order rungs; verified)**.** *Under `XiToeplitzTotalPositive`: (i) every 1×1 minor gives $\mu_n \geq 0$; (ii) the contiguous 2×2 minor equals the Turán determinant, $\mu_{n+1}^2 - \mu_n \mu_{n+2}$, so its nonnegativity is exactly the Turán inequality $b_n b_{n+2} \leq b_{n+1}^2$; (iii) for every order k and offset m the contiguous minor* $\det[\mu_{m+i-j}]_{i,j<k} \geq 0$.

We further prove (Lemma `xiToeplitzMinor3_sign_eq`) that the order-3 signed minor equals $(-1)^n$ times the unsigned b-determinant, exhibiting the correct alternating-positivity structure; and we document that the Hankel/moment route used in an earlier draft is the *wrong* machine, since Hankel positivity forces *log-convexity*, the opposite of the (true) Turán log-concavity.

3.2 Tightness of the reduction

Let the three classical external inputs be: `XiToeplitzTPToScaledFiniteBridge` (Edrei–ASW), `XiLaguerrePolyaClosureBridge` (LP closure), and `PolyaJensenBridge`; and let `XiToeplitzTPFromRH` be the classical converse (RH $\Rightarrow$ ξ is LP $\Rightarrow$ its coefficients are PF).

Theorem 5 (`xiToeplitzTotalPositive_iff_riemannHypothesis`; verified)**.** *Given the three classical bridges and the classical converse,*

$$\textit{XiToeplitzTotalPositive} \iff \text{RH}.$$

Remark 6 (The honesty linchpin)*.* Theorem 5 is the classical equivalence [3, 4, 5, 13] in machine-checked form. Its value is not mathematical novelty. It mechanically pins down the one hypothesis that is not classical, and shows that this input is *equivalent* to RH. Hence declaring it an axiom is logically identical to `axiom rh : RiemannHypothesis`: it does not constitute a proof, and `#print axioms` would expose the assumption. A reduction is progress only if its hypothesis is *weaker than, or independent of*, RH; this one is provably neither.

3.3 The complete moment-determinant ladder

Under Pólya's representation (the named external input `XiMomentKernelRep`, encoding (3)), the PF condition transfers to explicit moment determinants at *every* order.

Theorem 7 (`xiToeplitzContigMinor_eq_of_kernelRep`; verified)**.** *For all $k, m \in \mathbb{N}$,*

$$\det\big[\mu_{m+i-j}\big]_{i,j<k} \;=\; 2^k \cdot \det\left[\frac{M_{m+i-j}}{(2(m+i-j))!}\right]_{i,j<k}.$$

Consequently (`xiToeplitzContigMinor_nonneg_iff_moment_of_kernelRep`) Pólya-frequency positivity at all orders is equivalent to positivity of the factorial-weighted moment determinants of the explicit positive kernel Φ.

Proof (mechanized). Each entry of the signed Toeplitz matrix equals 2 times the corresponding factorial-weighted moment entry (by (3)); the determinant is then 2^k times the moment determinant by multilinearity (`Matrix.det_smul`, $\#\operatorname{Fin} k = k$). Positivity of 2^k gives the equivalence. □

The order-2 mechanism is quantified by the *slack factor* $s_n = \dfrac{(2n+3)(2n+4)}{(2n+1)(2n+2)}$, for which we verify $s_n > 1$ (`one_lt_slackFactor`) and $s_{n+1} < s_n$ (`slackFactor_succ_lt`); thus $s_n \downarrow 1$.

3.4 Order-2 unification: Turán ⇒ Jensen hyperbolicity

Theorem 8 (`jensenPoly_two_hyperbolic_of_turan`; verified)**.** *If $b_{n+2} \neq 0$ and the Turán inequality $b_n b_{n+2} \leq b_{n+1}^2$ holds, then the degree-2 Jensen polynomial $J_{2,n}$ is hyperbolic (all complex roots real). Hence (`allJensenTwo_hyperbolic_of_turan2All`) if all coefficients are nonzero and all Turán inequalities hold, every $J_{2,n}$ is hyperbolic.*

Proof (mechanized). $J_{2,n}$ maps to $b_{n+2}z^2 + 2b_{n+1}z + b_n$ over $\mathbb{C}$. Completing the square, $(b_{n+2}z + b_{n+1})^2 = b_{n+1}^2 - b_n b_{n+2} =: D \geq 0$ at any root. The auxiliary lemma `im_zero_of_sq_eq_nonneg_real` ($w^2 = D \in \mathbb{R}_{\geq 0} \Rightarrow \operatorname{Im} w = 0$) forces $b_{n+2}z + b_{n+1} \in \mathbb{R}$, hence (as $b_{n+2} \neq 0$ real) $\operatorname{Im} z = 0$. No square roots and no zero-locus input are used. □

Nondegeneracy $b_{n+2} \neq 0$ is supplied by (3) ($M_n > 0$). The analytic input is classical. Csordas, Norfolk and Varga [7] proved Pólya's conjectured Turán inequalities for the ξ coefficients. Dimitrov and Lucas [14] proved the higher-order (degree-3) Turán inequalities without assuming RH. Theorem 8 is the formal bridge from such inequalities to hyperbolicity, not a new inequality.

3.5 An independent operator-positivity route

The naive Hilbert–Pólya operator must be self-adjoint for the ambient inner product—the obstruction. The pseudo-Hermitian viewpoint [10, 11] keeps only a *positive metric*.

Definition 9 (`PseudoHermitianForm`)**.** For a $\mathbb{C}$-linear T on a $\mathbb{C}$-vector space E, a pseudo-Hermitian form is a map $\eta\colon E \times E \to \mathbb{C}$, linear in the first slot and conjugate-linear in the second, with $\eta(Tx, y) = \eta(x, Ty)$ (η-self-adjointness) and $\eta(x, x) \neq 0$ for $x \neq 0$ (nondegeneracy, supplied by positive-definiteness).

Theorem 10 (`PseudoHermitianForm.eigenvalue_conj_eq` and `riemannHypothesis_of_pseudoHermitianWitness`; verified)**.** *An η-self-adjoint operator has real eigenvalues. Consequently, if the critical-strip zeros are realized as the η-self-adjoint spectrum via the encoding $\mu(s) = -i\,(s - \frac{1}{2})$ (real iff $\operatorname{Re} s = \frac{1}{2}$), then RH holds.*

Proof (mechanized). For an eigenpair $Tv = \mu v$, $v \neq 0$: $\mu\,\eta(v, v) = \eta(Tv, v) = \eta(v, Tv) = \bar{\mu}\,\eta(v, v)$, and $\eta(v, v) \neq 0$, so $\mu = \bar{\mu}$. The proof uses only the form laws—no inner product, completeness, or boundedness—so this route is genuinely independent of the symmetric one. Existence of (E, T, η) with positive η and the prescribed spectrum is open. □

4 Numerical consistency checks

Coefficient enclosures. The coefficients $\mu_0, \ldots, \mu_{120}$ are enclosed by rigorous balls computed with FLINT/Arb [23] (python-flint 0.8.0, 4096 working bits). The computation uses series for the special functions in ξ, with explicit analytic remainders. Every enclosure has

relative accuracy of at least 2317 bits and exact rational endpoints. The first nine coefficients agree with an independent theta-integral enclosure. Data and generator are in `research/certified_xi/coefficients_120.json` (SHA-256 `adf3418d...63b11`) of the archived release. The June draft instead used two floating-point Cauchy-contour computations (620 and 520 digits), which agree to ≥ 236 digits. That is a consistency check, not an enclosure, and is superseded here.

Certified claims. The script `paper/certify_reduction_minors.py` works at 8192 bits. It certifies in ball arithmetic, so that each output ball lies strictly in $(0, \infty)$:

(i) $\mu_n > 0$ for $0 \leq n \leq 120$;

(ii) strict log-concavity $\mu_{n+1}^2 > \mu_n \mu_{n+2}$ for $0 \leq n \leq 118$;

(iii) every contiguous zero-extended Toeplitz minor $\det[\mu_{m+i-j}]_{i,j<k}$ with $1 \leq k \leq 16$ and $m + k - 1 \leq 120$ is strictly positive. These are 1816 minors, and the largest relative radius of a determinant ball is 6.5×10^{-669}, at $k = 16$, $m = 105$.

The certificate output, including the input hash, is stored in `paper/rh_reduction_minor_certificate.json`. The trust boundary is FLINT/Arb together with the classical identities behind the enclosures. These are not Lean certificates. Claim (iv) of the June draft, Jensen hyperbolicity for $d \leq 6$ and $n \leq 114$, was a floating-point root computation and remains uncertified. We report it only as a floating-point observation.

What the checks do and do not show. Nonnegativity in (iii) is already a theorem: Katkova's finite-order result covers every minor of order ≤ 43, at every index [13]. Claim (ii) is a special case of the infinite log-concavity of (μ_n) proved in [20]. That proof is a preprint and is computer-assisted. For Jensen polynomials, hyperbolicity is known for $d \leq 3$ at every shift [7, 14], and for every fixed d at all sufficiently large shifts [8, 15]. The computations above are therefore calibration of conventions and margins: strictness with explicit error control in a finite range. They are not discoveries of previously unknown positivity. The figures below were produced in June from the floating-point coefficients. These agree with the midpoints of the certified balls to relative error below 5×10^{-115} for every $n \leq 120$, far below plotting resolution.

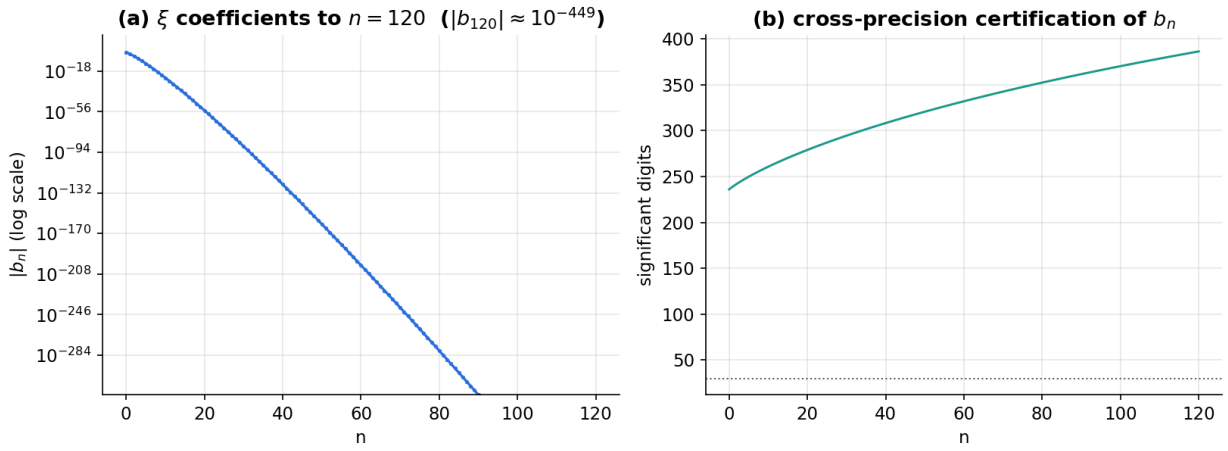


Figure 1: ξ Taylor data to record depth. (a) $|b_n|$ for $n \le 120$ on a log scale, decaying factorially to $|b_{120}| \approx 10^{-449}$; the alternating sign of b_n gives $\mu_n = (-1)^n b_n > 0$ (order-1 minors, forced by $\Phi > 0$). (b) June 2026 cross-precision consistency check: every b_n agrees to ≥ 236 significant digits between two floating-point computations. This check is superseded by the ball enclosures of Section 4.

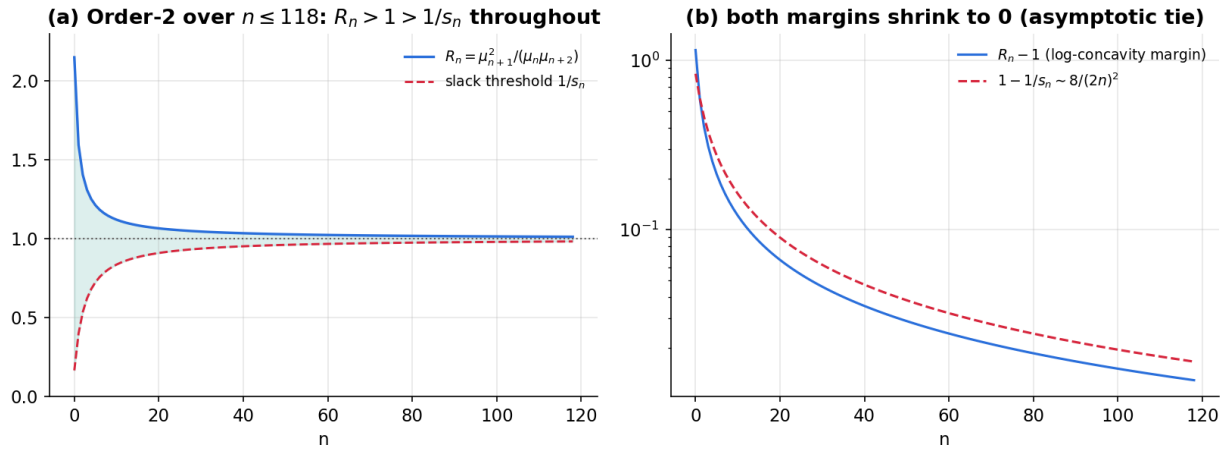


Figure 2: Order-2 over the full range $n \le 118$. (a) $R_n = \mu_{n+1}^2/(\mu_n\mu_{n+2}) > 1$ throughout (μ is genuinely log-concave—stronger than the slack requires) and stays above the slack threshold $1/s_n$ (shaded margin); both $R_n \to 1^+$ and $1/s_n \to 1^-$. (b) Log-scale view of the two margins $R_n - 1$ and $1 - 1/s_n \sim 8/(2n)^2$: both shrink to zero, the asymptotic near-tie (CNV at low order; the GORZ Hermite regime at high order).

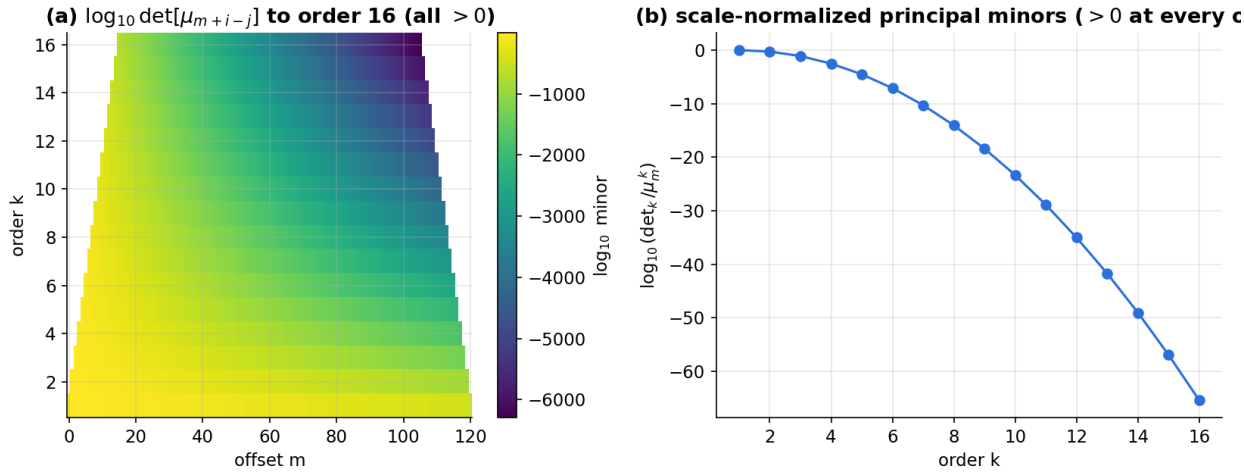


Figure 3: The Pólya-frequency ladder at scale. (a) $\log_{10}\det[\mu_{m+i-j}]$ for orders $k = 1, \ldots, 16$ and all offsets m (values span down to $\sim 10^{-6000}$). *Every minor is strictly positive*, as certified in ball arithmetic (Section 4); the smallness is factorial scale, not near-degeneracy. (b) Scale-normalized principal minors $\det_k / \mu_m^k$ remain positive at every order up to 16.

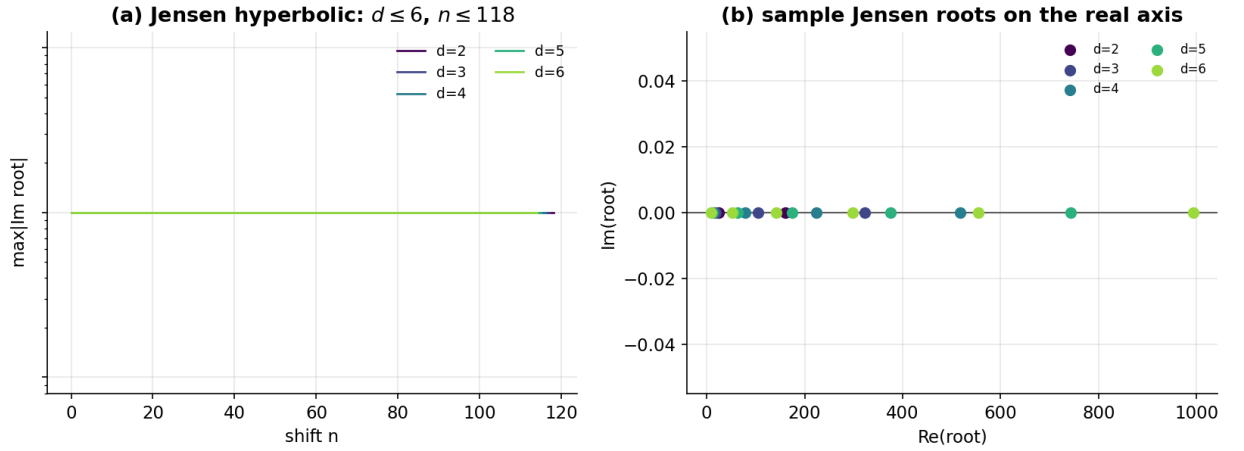


Figure 4: Jensen hyperbolicity at scale. (a) $\max|\operatorname{Im}(\text{root})|$ of $J_{d,n}$ for degrees $d = 2, \ldots, 6$ is numerically zero ($\sim 10^{-320}$) across all computed n up to 114. (b) Sample Jensen roots lie on the real axis. These are floating-point observations, not certified.

Interpretation. The reduction's premise holds for ξ at every tested order (Fig. 3), with comfortable low-order margins (Fig. 2). As the known theorems predict, μ is log-concave, which is strictly stronger than the slack-relaxed Turán requirement. Any fixed low rung is settled. The difficulty is *uniformity*: behaviour as $k \to \infty$, together with the shifts not covered by [18].

5 The open frontier

The reductions convert "zeros on a line" into positivity statements about the explicit positive kernel Φ and an explicit operator metric—objects amenable to tools foreign to zero-counting.

A. Uniform total positivity. Prove $\det[\mu_{m+i-j}] \geq 0$ for all k, m. This is RH-equivalent, by Theorem 5. Every *fixed* order up to 43 is classical [13]. A recent preprint claims strict positivity of every contiguous minor of order $r \geq 2$ at all shifts $m \geq 10^{18} r^3$, using certified saddle-point estimates [18]. The open content is therefore the complementary region of unbounded order and moderate shift. The continuous de Bruijn–Newman kernel is not PF_5 [19], so a continuous-kernel total-positivity shortcut is unavailable. Classical tools are Karlin total positivity [6], Schoenberg Pólya-frequency functions, and the variation-diminishing property of the kernel in $\sum \mu_n z^n = \int_0^\infty \Phi(u) \cosh(u\sqrt{z})\, du$.

B. Asymptotic margin. Control $R_n - 1$ versus $1 - 1/s_n \sim 8/(2n)^2$. Tools: the Griffin–Ono–Rolen–Zagier Hermite asymptotics [8] and their effective form [15]. Farmer argues that Jensen-polynomial asymptotics alone are not a plausible route to RH [16].

C. μ log-concavity (settled). The June draft listed $\mu_{n+1}^2 > \mu_n \mu_{n+2}$ as open. By (3) it is equivalent to $M_{n+1}^2/(M_n M_{n+2}) > 1/s_n = \frac{(2n+1)(2n+2)}{(2n+3)(2n+4)}$. The Turán inequalities of Csordas–Norfolk–Varga [7] give $M_{n+1}^2 > \frac{2n+1}{2n+3} M_n M_{n+2}$, which is stronger. A recent preprint proves much more, strict *infinite* log-concavity of (μ_n) [20]. Like Jensen asymptotics, this is a consequence of RH and does not imply it.

D. Operator existence. Construct (E, T, η) with a positive metric and spectrum $\{\gamma_k\}$ (operator theory; cf. [11]).

The de Bruijn–Newman programme provides an independent dynamical handle on the same object. It is known that $\Lambda \geq 0$ [9] and $\Lambda \leq 0.2$ [17], while RH $\iff \Lambda \leq 0$.

Remark 11 (A conditional finite-order improvement). Katkova's order 43 uses two inputs. The first is that nontrivial zeros satisfy $|\operatorname{Re}\rho - \frac{1}{2}| \leq \frac{1}{2}$. The second is that no zero has $|\operatorname{Im}\rho| < \gamma_1 = 14.1347\ldots$. In the variable $z = (s - \frac{1}{2})^2$, a zero $\frac{1}{2} + x + i\gamma$ lies at angle $2\arctan(|x|/\gamma)$ from the negative axis. Schoenberg's sector theorem yields PF_m when this angle is at most $\pi/(m+1)$. A recent manuscript claims that $\zeta(s) \neq 0$ for $\operatorname{Re} s > 7/8$ [21]. Its statement is machine-checked in an external Lean library against Mathlib's `riemannZeta`, but it has not yet been refereed. If the claim holds, then $|x| \leq \frac{3}{8}$, the maximal angle drops from 0.07072 to 0.05305, and the same argument gives PF_{58} in place of PF_{43}. This is a constant improvement only. Every fixed order already follows from RH verified to a sufficient height, so fixed order is not RH-strength. The remark is not used elsewhere in this paper.

6 Methodology, reproducibility, and limitations

Verification. Every Lean theorem and named input cited in this paper exists at commit `c749786` of `https://github.com/sciencenerd-des/reinmann`. That commit is archived with DOI 10.5281/zenodo.23187646 [24]. It uses Lean `v4.30.0` and Mathlib `v4.30.0`, with 78 Lean files and entry point `Reinmann.lean`. The development has grown well beyond the 47 modules of the June draft; this paper uses only the results cited here. On 7 October 2026 we rebuilt that commit and ran `#print axioms` on each cited theorem; every footprint is {propext, Classical.choice, Quot.sound}. The repository scripts `scripts/safe-verify.sh` and `scripts/verify_axiom_clean.sh` reject `sorry`, `admit`, added axioms and similar shortcuts.

Named external inputs (never asserted). `PolyaJensenBridge`, `XiToeplitzTPToScaledFiniteBridge`, `XiLaguerrePolyaClosureBridge`, `XiMomentKernelRep`,

`XiToeplitzTPFromRH`. Each is a standard classical theorem; formalizing them in Mathlib is future work.

Limitations.

1. No proof of RH: every RH-implying statement is conditional on an input that is RH-equivalent (Theorem 5) or an unformalized classical theorem.
2. The numerics are finite. Ball arithmetic certifies positivity only for $n \leq 120$ and Toeplitz order $k \leq 16$, under the FLINT/Arb trust boundary. The Jensen observations for $d \leq 6$ are floating-point only. None of this can establish the uniform $(k, n \to \infty)$ statement that is RH-equivalent, and the certified range lies inside what known theorems already cover.
3. The pseudo-Hermitian existence claim is entirely open and unevidenced.
4. Equivalence-chain reformulations (e.g. "≤ 1 zero per height") are correct but RH-equivalent, hence reorganization rather than progress.

7 Related work

Classical background. Classical sources are Pólya [2, 3], Aissen–Schoenberg–Whitney [4], Edrei [5] and Karlin [6]. For the ξ coefficients specifically, Katkova [13] gives the Pólya-frequency formulation and finite-order positivity. The Turán and higher-order Turán inequalities are due to Csordas–Norfolk–Varga [7] and Dimitrov–Lucas [14].

Jensen polynomials and heat flow. Jensen-polynomial asymptotics are developed in [8, 15], with the critique of [16]. On the de Bruijn–Newman constant see [9, 17].

Recent work on the same coefficients. Three 2026 preprints concern exactly the coefficients studied here: a tail wedge for contiguous Toeplitz minors [18], certified failure of PF_5 for the continuous kernel [19], and infinite log-concavity [20]. A claimed zero-free half-plane $\mathrm{Re}\, s > 7/8$ [21] bears on finite-order constants only (Remark 11).

Operators and formalization. For the pseudo-Hermitian viewpoint see Mostafazadeh [10] and Bender–Brody–Müller [11]. The formal foundations are Mathlib [12] and its zeta and L-function library [22]. Spectral and adelic perspectives (Berry–Keating, Connes) and the Weil and Li positivity criteria are classical antecedents of the operator and positivity viewpoints used here.

Acknowledgement of limitations (summary)

The verified contribution is a correct, tight, axiom-clean *reduction* of RH to an explicit positivity statement, plus an independent operator reduction, a repaired target-function error, and numerical evidence. RH itself remains open, as stated throughout.

Disclosure of AI-assisted tools

In preparing this work the author used large language model assistants—OpenAI GPT-5.5 and Anthropic Claude Opus 4.8—for drafting and editing of exposition, literature triage, code scaffolding for the numerical scripts, and consistency checking of the manuscript. These tools were used strictly as assistance under the author's direction. They are not authors and bear no responsibility for the work: all mathematical statements, the Lean 4 formalization and its axiom audit, the high-precision numerical computations, and the final text were specified, checked, and verified by the author, who takes full and sole responsibility for the content, including any errors. No claim of correctness rests on the AI tools; every "verified" result is established by the Lean kernel and the `scripts/safe-verify.sh` discipline as described in the text.

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